

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

DDL

1) Create admin table

```
mysql> CREATE TABLE admin (  
->     adminID INT AUTO_INCREMENT PRIMARY KEY,  
->     name VARCHAR(100) NOT NULL,  
->     ic_no VARCHAR(20) NOT NULL UNIQUE,  
->     password VARCHAR(50) NOT NULL,  
->     last_login TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
SELECT * FROM `admin`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

adminID	name	ic_no	password	last_login
---------	------	-------	----------	------------

-Create loanPolicy table

```
mysql> CREATE TABLE loanPolicy (  
->     loanPolicyID INT AUTO_INCREMENT PRIMARY KEY,  
->     policyName VARCHAR(50) NOT NULL,  
->     loanRate DECIMAL(5, 2) NOT NULL,  
->     loanType ENUM('Pembiayaan_Skim_Khas', 'Pembiayaan_Al_Bai', 'Pembiayaan_RoadTaxInsuran', 'Pembiayaan_Al_Innah') NOT NULL,  
->     eligibility_criteria TEXT NOT NULL,  
->     last_update TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
->     last_updated_by INT NOT NULL,  
->     FOREIGN KEY (last_updated_by) REFERENCES admin(adminID)  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
SELECT * FROM `loanpolicy`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	last_updated_by
--------------	------------	----------	----------	----------------------	-------------	-----------------

-Create loanRateHistory table

```
mysql> CREATE TABLE loanRateHistory (  
->     loanRateHistory_id INT AUTO_INCREMENT PRIMARY KEY,  
->     loanPolicyID INT NOT NULL,  
->     currentRate DECIMAL(5, 2) NOT NULL,  
->     newRate DECIMAL(5, 2) NOT NULL,  
->     changeAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
->     updatedBy INT NOT NULL,  
->     FOREIGN KEY (loanPolicyID) REFERENCES loanPolicy(loanPolicyID),  
->     FOREIGN KEY (updatedBy) REFERENCES admin(adminID)  
-> );
```

```
SELECT * FROM `loanratehistory`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
--------------------	--------------	-------------	---------	----------	-----------

-Create member table

```
mysql> CREATE TABLE member (
->   memberID INT AUTO_INCREMENT PRIMARY KEY,
->   full_name VARCHAR(225) NOT NULL,
->   ic_no VARCHAR(20) NOT NULL UNIQUE,
->   gender ENUM('Male', 'Female') NOT NULL,
->   religion VARCHAR(20) NOT NULL,
->   race VARCHAR(20) NOT NULL,
->   marital_status ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL,
->   m_pf_number VARCHAR(20) NOT NULL,
->   position VARCHAR(100) NOT NULL,
->   grade VARCHAR(20),
->   monthly_salary DECIMAL(10, 2) NOT NULL,
->   home_address TEXT NOT NULL,
->   home_postcode VARCHAR(10) NOT NULL,
->   home_state VARCHAR(10) NOT NULL,
->   phone_no VARCHAR(20) NOT NULL,
->   status ENUM('Active', 'Inactive') NOT NULL DEFAULT 'Active'
-> );
Query OK, 0 rows affected (0.06 sec)
```

[SELECT * FROM `member`](#)

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

member_id	full_name	ic_no	gender	religion	race	marital_status	m_pf_number	position	grade	monthly_salary	home_address	home_postcode	home_state	phone_no	status
-----------	-----------	-------	--------	----------	------	----------------	-------------	----------	-------	----------------	--------------	---------------	------------	----------	--------

2)Alter member table

```
mysql> ALTER TABLE member MODIFY member_id INT NOT NULL AUTO_INCREMENT;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

[SELECT * FROM `member`](#)

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

member_id	full_name	ic_no	gender	religion	race	marital_status	m_pf_number	position	grade	monthly_salary	home_address	home_postcode	home_state	phone_no	status	managed_by_adminID
-----------	-----------	-------	--------	----------	------	----------------	-------------	----------	-------	----------------	--------------	---------------	------------	----------	--------	--------------------

DML

3) Insert data into admin table

```
mysql> INSERT INTO admin (name, ic_no, password)
-> VALUES
-> ('ZURAINI BINTI MOHAMAD ALI', '770506080456', 'zuraini@123'),
-> ('AHMAD AYDEN BIN JOHARI', '720703020877', 'ahmad@134');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

✓ Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

`SELECT * FROM `admin``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

		adminID	name	ic_no	password	last_login
☐	Edit Copy Delete	1	ZURAINI BINTI MOHAMAD ALI	770506080456	zuraini@123	2025-01-24 12:53:22
☐	Edit Copy Delete	2	AHMAD AYDEN BIN JOHARI	720703020877	ahmad@134	2025-01-24 12:53:22

4) Insert data into member table

```
mysql> INSERT INTO member
-> (
->     full_name, ic_no, gender, religion, race, marital_status,
->     m_pf_number, position, grade, monthly_salary,
->     home_address, home_postcode, home_state, phone_no, managed_by_adminID
-> )
-> VALUES
-> ('ALICE TAN', '890101023456', 'Female', 'Buddhism', 'Chinese', 'Single',
-> 'MPF789', 'Manager', '55', 9000.00, '45, JALAN BINTANG', '81310', 'Johor', '0187654321', 1),
->
-> ('MOHAMAD ALI', '760505012222', 'Male', 'Islam', 'Malay', 'Married',
-> 'MPF987', 'Supervisor', '50', 4800.00, '33, JALAN KUALA', '81300', 'Johor', '0191234567', 2),
->
-> ('NURUL HUDA', '950303011234', 'Female', 'Islam', 'Malay', 'Married',
-> 'MPF654', 'Technician', '52', 4000.00, '25, JALAN MERDEKA', '81320', 'Johor', '0172345678', 1),
->
-> ('CALYN LIM', '740601012222', 'Female', 'Christianity', 'Chinese', 'Married',
-> 'MPF123', 'Accountant', '53', 7500.00, '12, JALAN CAHAYA', '81300', 'Johor', '0173457789', 1),
->
-> ('ZULKIFLI BIN MOHAMAD NAZMI', '870412010999', 'Male', 'Islam', 'Malay', 'Single',
-> 'MPF456', 'Executive', '51', 5500.00, '19, JALAN KEBANGSAAN, TAMAN UNIVERSITI', '81300', 'Johor', '0198765432', 2);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

memberID	full_name	ic_no	gender	religion	race	marital_status	m_pf_number	position	grade	monthly_salary	home_address	home_postcode	home_state	phone_no	status	managed_by_adminID
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	MPF789	Manager	55	9000.00	45, JALAN BINTANG	81310	Johor	0187654321	Active	1
2	MOHAMAD ALI	760505012222	Male	Islam	Malay	Married	MPF987	Supervisor	50	4800.00	33, JALAN KUALA	81300	Johor	0191234567	Active	2
3	NURUL HUDA	950303011234	Female	Islam	Malay	Married	MPF654	Technician	52	4000.00	25, JALAN MERDEKA	81320	Johor	0172345678	Active	1
4	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	MPF123	Accountant	53	7500.00	12, JALAN CAHAYA	81300	Johor	0173457789	Active	1
5	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	MPF456	Executive	51	5500.00	19, JALAN KEBANGSAAN, TAMAN UNIVERSITI	81300	Johor	0198765432	Active	2

5) Make member status become inactive

```
mysql> UPDATE member
-> SET status = 'Inactive'
-> WHERE memberID = 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

memberID	full_name	ic_no	gender	religion	race	marital_status	m_pf_number	position	grade	monthly_salary	home_address	home_postcode	home_state	phone_no	status	managed_by_adminID
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	MPF789	Manager	55	9000.00	45, JALAN BINTANG	81310	Johor	0187654321	Inactive	1
2	MOHAMAD ALI	760505012222	Male	Islam	Malay	Married	MPF987	Supervisor	50	4800.00	33, JALAN KUALA	81300	Johor	0191234567	Active	2
3	NURUL HUDA	950303011234	Female	Islam	Malay	Married	MPF654	Technician	52	4000.00	25, JALAN MERDEKA	81320	Johor	0172345678	Active	1
4	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	MPF123	Accountant	53	7500.00	12, JALAN CAHAYA	81300	Johor	0173457789	Active	1
5	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	MPF456	Executive	51	5500.00	19, JALAN KEBANGSAAN, TAMAN UNIVERSITI	81300	Johor	0198765432	Active	2

6) Insert data into loanPolicy table

```
mysql> INSERT INTO loanPolicy (policyName, loanRate, loanType, eligibility_criteria, last_updated_by)
-> VALUES
-> ('Pembiayaan Skim Khas', 3.50, 'Pembiayaan_Skim_Khas', 'Minimum salary of RM3000', 1),
-> ('Pembiayaan_RoadTaxInsuran', 5.00, 'Pembiayaan_RoadTaxInsuran', 'Only available for government employees', 2);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

	loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	last_updated_by
☐ Edit ☐ Copy ☐ Delete	1	Pembiayaan Skim Khas	3.50	Pembiayaan_Skim_Khas	Minimum salary of RM3000	2025-01-25 17:25:26	1
☐ Edit ☐ Copy ☐ Delete	2	Pembiayaan_RoadTaxInsuran	5.00	Pembiayaan_RoadTaxInsuran	Only available for government employees	2025-01-25 17:25:26	2

7) Update loan Policy to modify loan rate or loan eligibility criteria

Update Query to modify the loan rate

```
mysql> UPDATE loanPolicy
-> SET loanRate = 4.20
-> WHERE policyName = 'Pembiayaan Skim Khas';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	last_updated_by
1	Pembiayaan Skim Khas	4.20	Pembiayaan_Skim_Khas	Minimum salary of RM3000	2025-01-25 20:14:01	1
2	Pembiayaan_RoadTaxInsuran	5.00	Pembiayaan_RoadTaxInsuran	Only available for government employees	2025-01-25 17:25:26	2

Update Query to modify the loan eligibility criteria

```
mysql> UPDATE loanPolicy
-> SET eligibility_criteria = 'Only available for kada cooperative's member'
-> WHERE policyName = 'Pembiayaan_RoadTaxInsuran';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	last_updated_by
1	Pembiayaan Skim Khas	4.20	Pembiayaan_Skim_Khas	Minimum salary of RM3000	2025-01-25 20:14:01	1
2	Pembiayaan_RoadTaxInsuran	5.00	Pembiayaan_RoadTaxInsuran	Only available for kada cooperative's member	2025-01-25 20:30:35	2

8) Insert data into loanRateHistory

```
mysql> INSERT INTO loanRateHistory (loanPolicyID, currentRate, newRate, updatedBy)
-> VALUES
-> ((SELECT loanPolicyID FROM loanPolicy WHERE policyName = 'Pembiayaan Skim Khas'), 3.50, 4.20, 1);
Query OK, 1 row affected (0.01 sec)
```

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
1	1	3.50	4.20	2025-01-25 20:21:05	1

10)Retrieve Loan Policy and Admin Info to show which admin update which loan policy

```

SELECT
    loanPolicy.loanPolicyID,
    loanPolicy.policyName,
    loanPolicy.loanRate,
    loanPolicy.loanType,
    loanPolicy.eligibility_criteria,
    loanPolicy.last_update,
    admin.adminID,
    admin.name AS AdminName
FROM
    loanPolicy
JOIN
    admin
ON
    loanPolicy.last_updated_by = admin.adminID;

```

```

mysql> SELECT
-> loanPolicy.loanPolicyID,
-> loanPolicy.policyName,
-> loanPolicy.loanRate,
-> loanPolicy.loanType,
-> loanPolicy.eligibility_criteria,
-> loanPolicy.last_update,
-> admin.adminID,
-> admin.name AS AdminName
-> FROM
-> loanPolicy
-> JOIN
-> admin
-> ON
-> loanPolicy.last_updated_by = admin.adminID;
+-----+-----+-----+-----+-----+-----+-----+
| loanPolicyID | policyName          | loanRate | loanType          | eligibility_criteria | last_update | adminID | AdminName          |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Pembiayaan Skim Khas | 4.20 | Pembiayaan_Skim_Khas | Minimum salary of RM3000 | 2025-01-25 20:14:01 | 1 | ZURAINI BINTO MOHAMAD ALI |
| 2 | Pembiayaan_RoadTaxInsuran | 5.00 | Pembiayaan_RoadTaxInsuran | Only available for kada cooperative's member | 2025-01-25 20:30:35 | 2 | AHMAD AYDEN BIN JOHARI |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	adminID	AdminName
1	Pembiayaan Skim Khas	4.20	Pembiayaan_Skim_Khas	Minimum salary of RM3000	2025-01-25 20:14:01	1	ZURAINI BINTO MOHAMAD ALI
2	Pembiayaan_RoadTaxInsuran	5.00	Pembiayaan_RoadTaxInsuran	Only available for kada cooperative's member	2025-01-25 20:30:35	2	AHMAD AYDEN BIN JOHARI

10)Join show which member manage by which admin

```
mysql> SELECT
->     m.ic_no AS member_ic,
->     m.full_name AS member_name,
->     a.name
-> FROM
->     member m
-> JOIN
->     admin a ON m.managed_by_adminID = a.adminID;
```

member_ic	member_name	name
890101023456	ALICE TAN	ZURAINI BINTO MOHAMAD ALI
760505012222	MOHAMAD ALI	AHMAD AYDEN BIN JOHARI
950303011234	NURUL HUDA	ZURAINI BINTO MOHAMAD ALI
740601012222	CALYN LIM	ZURAINI BINTO MOHAMAD ALI
870412010999	ZULKIFLI BIN MOHAMAD NAZMI	AHMAD AYDEN BIN JOHARI

5 rows in set (0.00 sec)

member_ic	member_name	admin_name
890101023456	ALICE TAN	ZURAINI BINTO MOHAMAD ALI
760505012222	MOHAMAD ALI	AHMAD AYDEN BIN JOHARI
950303011234	NURUL HUDA	ZURAINI BINTO MOHAMAD ALI
740601012222	CALYN LIM	ZURAINI BINTO MOHAMAD ALI
870412010999	ZULKIFLI BIN MOHAMAD NAZMI	AHMAD AYDEN BIN JOHARI

11) Filter Active Members or Inactive Member

Active

```
mysql> SELECT full_name,ic_no,position,status
-> FROM member
-> WHERE status ='Active';
```

full_name	ic_no	position	status
MOHAMAD ALI	760505012222	Supervisor	Active
NURUL HUDA	950303011234	Technician	Active
CALYN LIM	740601012222	Accountant	Active
ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Executive	Active

4 rows in set (0.00 sec)

Inactive

```
mysql> SELECT
-> full_name,ic_no,position,status
-> FROM
-> member
-> WHERE
-> status = 'Inactive';
```

full_name	ic_no	position	status
ALICE TAN	890101023456	Manager	Inactive

1 row in set (0.00 sec)

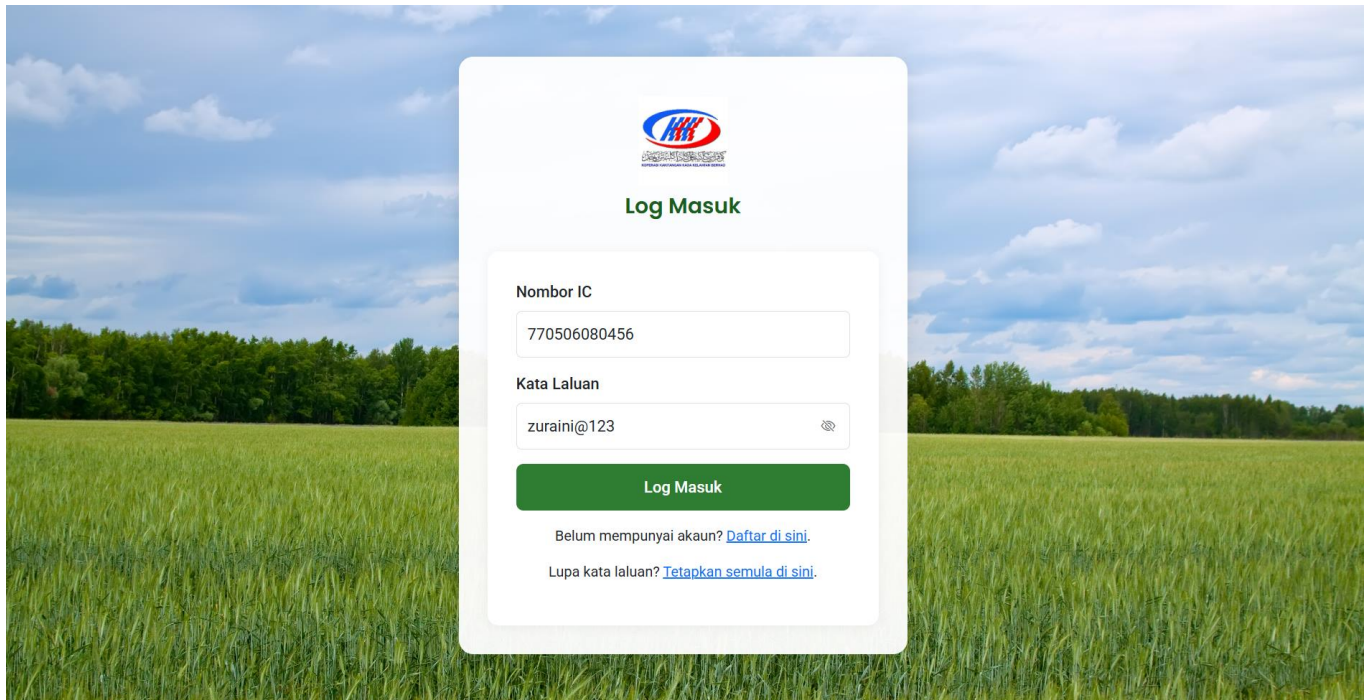
12)Remove member

```
mysql> DELETE FROM member
-> WHERE memberID = 2;
Query OK, 1 row affected (0.01 sec)
```

memberID	full_name	ic_no	gender	religion	race	marital_status	m_pf_number	position	grade	monthly_salary	home_address	home_postcode	home_state	phone_no	status	managed_by_adminID
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	MPF789	Manager	55	9000.00	45, JALAN BINTANG	81310	Johor	0187654321	Inactive	1
3	NURUL HUDA	950303011234	Female	Islam	Malay	Married	MPF654	Technician	52	4000.00	25, JALAN MERDEKA	81320	Johor	0172345678	Active	1
4	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	MPF123	Accountant	53	7500.00	12, JALAN CAHAYA	81300	Johor	0173457789	Active	1
5	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	MPF456	Executive	51	5500.00	19, JALAN KEBANGSAAN, TAMAN UNIVERSITI	81300	Johor	0198765432	Active	2

3.2 Interface Implementation

Admin can use their identification number and password to log into the system.



Admin can delete the member if the member wants to quit the cooperative. Admin can also delete the member that is inactive. Admin can also deactivate members. It provides a drop-down button so that the Admin can filter the dashboard won't be cluttered with much information.

Admin can deactivate the member

Member Table										
Filter by Status:										
All										
#	Full Name	IC No	Gender	Religion	Race	Marital Status	Position	Salary	Status	Actions
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	Manager	9000.00	Inactive	Activate Delete
2	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	Accountant	7500.00	Active	Deactivate Delete
3	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	Executive	5500.00	Active	Deactivate Delete
4	MOHAMAD ALI	760505012222	Male	Islam	Malay	Married	Supervisor	4800.00	Active	Deactivate Delete
5	NURUL HUDA	950303011234	Female	Islam	Malay	Married	Technician	4000.00	Active	Deactivate Delete

Filter by active member

Member Table										
Filter by Status:										
Active										
#	Full Name	IC No	Gender	Religion	Race	Marital Status	Position	Salary	Status	Actions
2	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	Accountant	7500.00	Active	Deactivate Delete
3	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	Executive	5500.00	Active	Deactivate Delete
4	MOHAMAD ALI	760505012222	Male	Islam	Malay	Married	Supervisor	4800.00	Active	Deactivate Delete
5	NURUL HUDA	950303011234	Female	Islam	Malay	Married	Technician	4000.00	Active	Deactivate Delete

Filter by inactive member

Member Table										
Filter by Status:										
Inactive										
#	Full Name	IC No	Gender	Religion	Race	Marital Status	Position	Salary	Status	Actions
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	Manager	9000.00	Inactive	Activate Delete

Delete member with memberID =2

Member Table											
Filter by Status:											
All											
#	Full Name	IC No	Gender	Religion	Race	Marital Status	Position	Salary	Status	Actions	
1	ALICE TAN	890101023456	Female	Buddhism	Chinese	Single	Manager	9000.00	Active	Deactivate	Delete
2	CALYN LIM	740601012222	Female	Christianity	Chinese	Married	Accountant	7500.00	Active	Deactivate	Delete
3	ZULKIFLI BIN MOHAMAD NAZMI	870412010999	Male	Islam	Malay	Single	Executive	5500.00	Active	Deactivate	Delete
4	MOHAMAD ALI	760505012222	Male	Islam	Malay	Married	Supervisor	4800.00	Active	Deactivate	Delete
5	NURUL HUDA	950303011234	Female	Islam	Malay	Married	Technician	4000.00	Active	Deactivate	Delete

Admin can update the loan policy and loan rate

The admin picks an existing loan policy and corresponding loan type to update.

The admin can update eligibility criteria or leave them unchanged.

Admin can update the loan rate. If updated, a record is saved in history.

Update Loan Policy button: Updates the policy (name, loan type, and criteria).

Update Loan Rate button: Updates only the loan rate.

Mandatory Fields: The admin must select a loan policy and loan type to submit the form.

Edit Loan Policy	
Select Loan Policy to Update	
Select a Loan Policy	
Please select the loan policy you wish to update. This field is required to proceed with any updates.	
Policy Name	
Leave empty to retain current policy name	
Update the policy name here. If you don't want to change it, leave this field empty.	
Loan Type	
Select Loan Type	
Select the type of loan for the policy update. This field is required.	
Eligibility Criteria	
Enter eligibility criteria (Leave empty to retain current criteria)	
Update the eligibility criteria here. If you don't wish to modify it, leave this field empty.	
Loan Rate	
Enter new loan rate (Leave empty to retain current rate)	
Enter a new loan rate if you wish to update it. If not, leave this field empty to maintain the current rate.	
Note: Changing the loan rate will generate a record in the loan rate history.	
Update Loan Policy	Update Loan Rate

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

The proposed administrative module significantly enhances the usability, efficiency, and flexibility of the system. This module addresses most of the key identified weaknesses, which were on the inability to manage loan rates without technical skills, update loan policies, and manage members' records without technical experience. It introduces a user-friendly interface through which administrators can now make timely modifications of loan rates and policy management without involving developers. It also helps to do the job in terms of membership account management; this can be taken further when taking away inactive members or when a member wants to quit the cooperative. This makes the admin's dashboard clean and neat for efficiency. Hence, the system can be more aligned with organizational needs, able to quickly and effectively react toward policy changes. This will provide better value for administrators as well as members. The modular design makes it scalable, reduces operational delays, and helps in a generally smooth workflow. Moving forward, the module could be further improved by the ability of the administrators to deactivate the members for specific reasons like quitting, passing away, or retiring. This will ensure that accurate records are retained and the dashboard is neat. Additionally, this will help in the generation and analysis of better reports on the trend and reasons behind the deactivation of members, which then informs the management of the cooperative in decision-making.

4.2 Reflection

The WBL project aimed at the development of a new system to replace the outdated manual system of Koperasi Kakitangan KADA Lembaga Kelantan Berhad. These manual processes were time-consuming, inefficient, and prone to errors, leading to delays and dissatisfaction among users. Our group proposed an integrated system to address these challenges, with a primary focus on automating and streamlining loan approvals, membership management, and saving account processes. I work on the administrative module, which is supposed to approve loans, manage memberships, handle transactions, reply to inquiries, and generate analysis reports. My contributions focus on ensuring these tasks are streamlined and efficient, providing administrators with the tools needed to manage operations effectively and meet user needs. Other members worked on other modules within the system: the Loan Application Module and User Authentication were by Elijah She Yu Sheng, Report Generating Module for members was done by Choh Jing Yi, Lee Yin Shen handled the Membership Application Module, while Lim Yu Han was in charge of the Savings Account Module. Our efforts, together, therefore, aimed to integrate this system in such a way that the requirements of users were comprehensively achieved.

At that time, I felt excited and challenged during this period. I was excited because I was dealing with a real-world issue and was able to create something tangible, but challenges also lay the coding errors on the administrative module, for one. These needed a great deal of effort and time to get right, adding to the pressure since we had very little time to present the task. Moreover, we were also balancing other courses, projects, assignments, and examinations, adding to the stress. The frustration of making mistakes brought to light the cautious debugging and testing of a program. There were discouraging moments, especially when some minor mistakes took so much time that was not anticipated, yet team support and our shared sense of purpose motivated me to push through. Moreover, the limitation of resources and time made us more constrained, which influenced the actions and behaviors of everyone as we worked under pressure to meet deadlines. Despite these difficulties, we managed to overcome the challenges together and achieve our objectives.

The WBL project was rewarding and challenging at the same time. It was exciting to work on a real-world problem and collaborate with a dedicated team. However, debugging coding errors and balancing the project with other courses, assignments, and exams was stressful. Time constraints and resource limitations added pressure, leading to discouragement when issues took longer to fix. One of the big challenges was that we finished some parts of the system, but to make it perfect, we wanted to add more or make it more functional, and then some new errors appeared; it took so much time to fix them. Our team also lacked careful planning in the early stage which not just caused us to revisit and redo parts of the system to enhance it, but we also wasted time in terms of clearly assigning responsibilities for each module. Instead of working in parallel, we started the modules one by one, which resulted in wasted time. Fortunately, we learned this valuable lesson early on and very swiftly adapted to these changes so we could work faster, the modification helped us move ahead while avoiding other snags. While team support helped, this project showed that better communication among its members is required. In the future, better planning, communication, and feedback will be instrumental in ensuring smoother collaboration and better results.

I have chosen to reflect on this incident because it represents some of the key challenges we faced during the WBL project related to debugging, time management, and communication. The effort of enhancing parts of the system and facing new errors made me understand the importance of managing changes and expectations properly. It makes sense given the previous stages, as we already faced stresses of balancing multitasking pressures and the scope of the project was just adding to that stress. The major area of improvement in this regard for the future is improving our team communication. Better communication would avoid unnecessary mistakes, and ensure smoother coordination, and better results. Additionally, better planning and a structured feedback loop would help manage system improvements without introducing new errors.

This experience taught me an important lesson in clear communication, careful planning, and thorough testing of our project. Even minor changes to the system could introduce new bugs, and debugging those took much time and required a team effort. The balancing act of assignments and exams also taught me how to prioritize and manage my time more effectively. Moving forward, I understand that proper collaboration and more structured communication will help to avoid unnecessary mistakes. However, the question would still be how to effectively streamline communication and get all team members on board along the process to ensure that rework and delays will be avoided.

Next time, I will prioritize in increasing communication and planning from early stage. I will make sure every team member understands their individual tasks for their segments, and we will work simultaneously from the start to avoid wasting time. Regular meetings and updates will keep us on course, catch any problems in advance, and implement necessary changes. I will also establish a more rigorous testing and debugging process to minimize the aggravation of constant errors. I will be employing VsCode Live stream pair programming and GitHub as our primary communication. This will enable us to monitor progress, handle version control, and ensure everyone knows what is going on. Furthermore, improved coding and documentation practices can prevent errors. Having others, such as colleagues or mentors, look over our work sooner in the process can also allow us to find errors when they are still small problems.